

# Bond disorder in non collinear antiferromagnets

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# Bond disordered Heisenberg magnets

$$H = \sum_{i,j} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$$



Disordered couplings

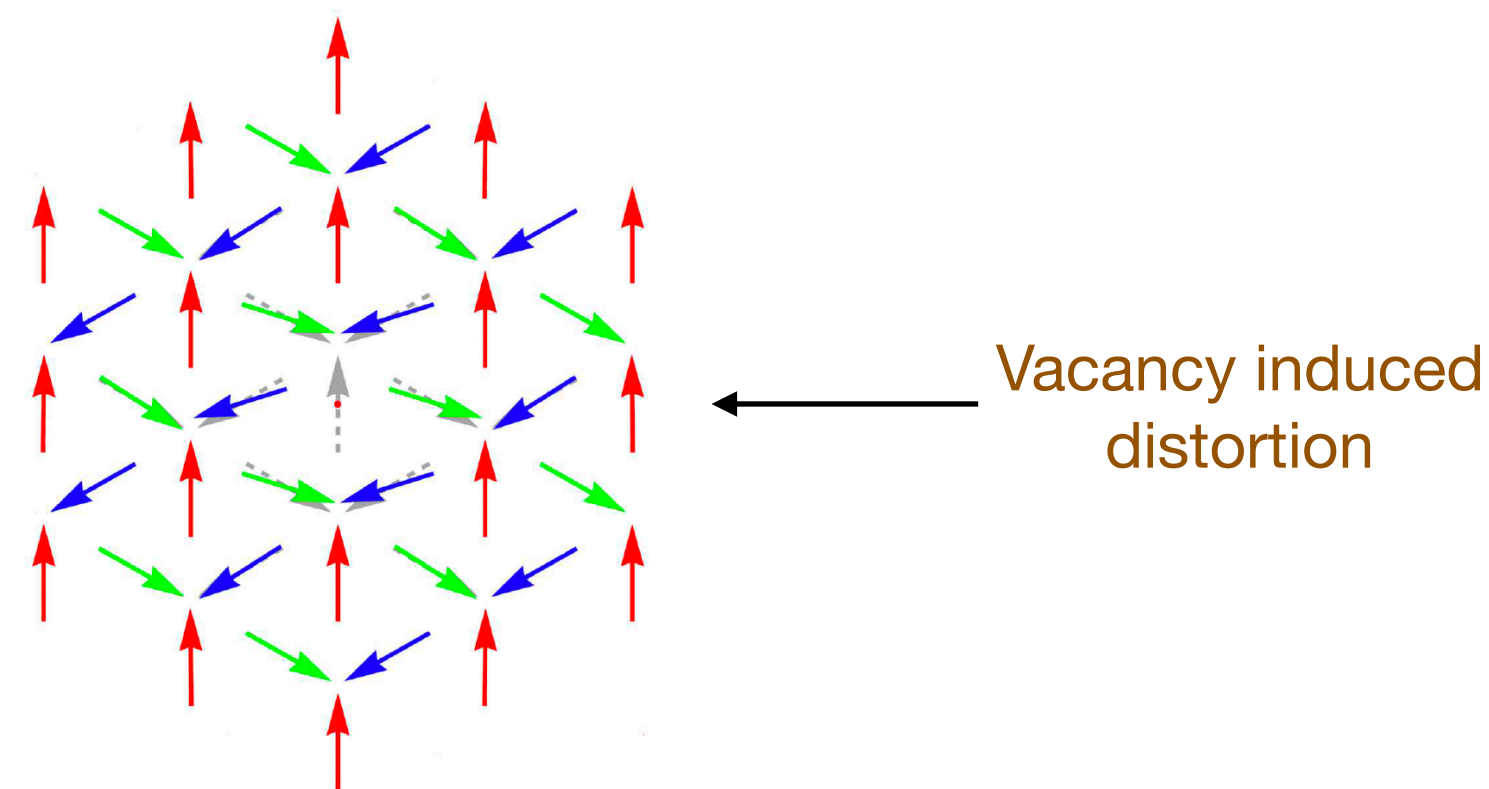
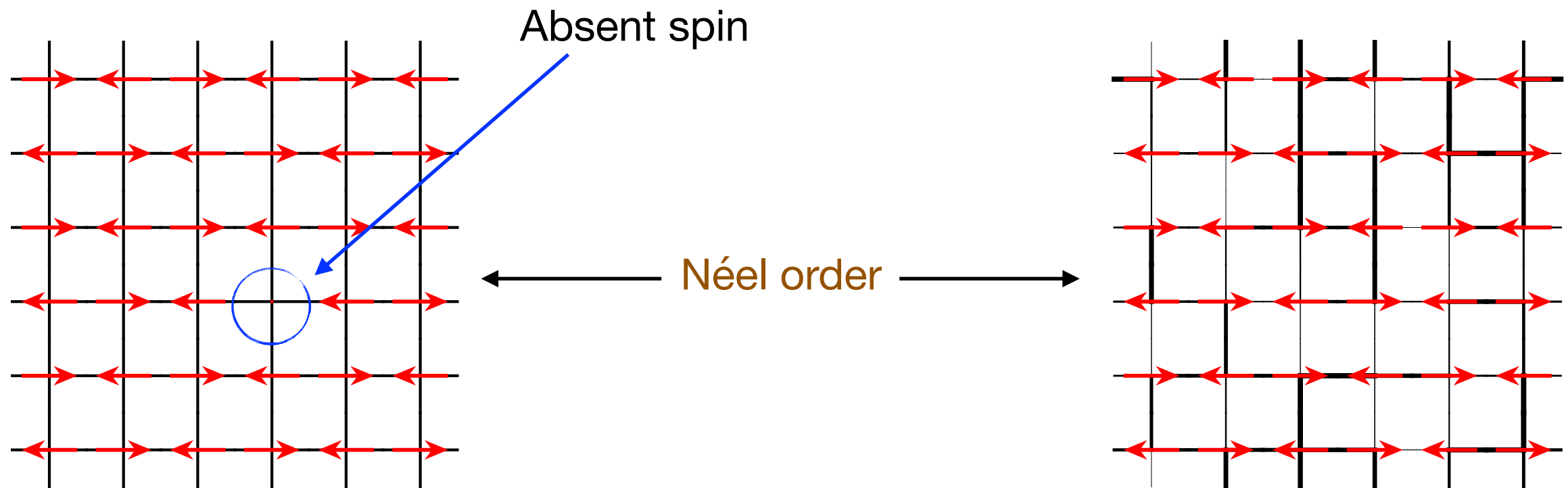
- Except from lattice translational invariance, they do not break any other symmetries of the Hamiltonian.
- However, bond disorder effects can be nontrivial for non collinear antiferromagnetic order.

Non collinear antiferromagnets are often more susceptible to the effects of defects and disorder.

# Heisenberg antiferromagnet

Vacancy

Bond disorder



Is the long range order (LRO) affected by bond disorder?

Heisenberg  
antiferromagnet  
(continuous symmetry)  
with non collinear LRO

Bond disorder

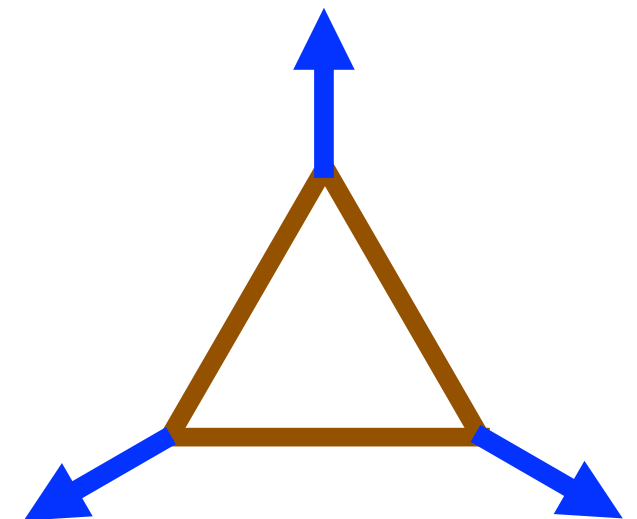


The fate of LRO?

Triangular lattice Heisenberg antiferromagnet

$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j$$

Uniform  $J_{ij} = J$



120 degree order

Spatially random  $J_{ij}$



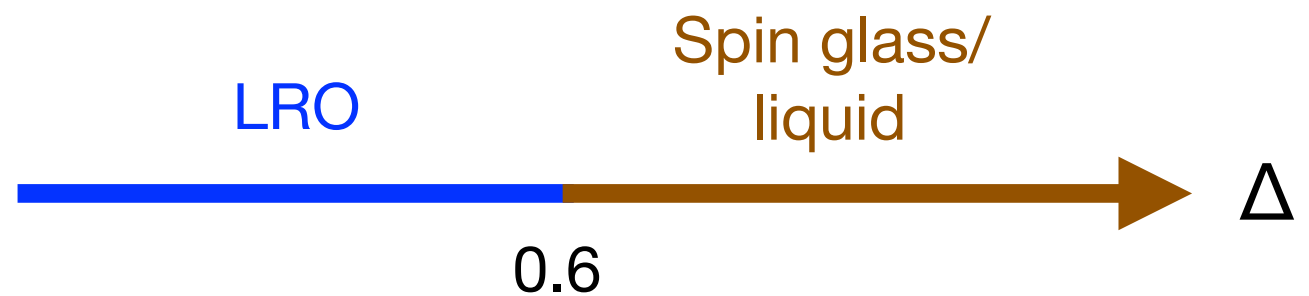
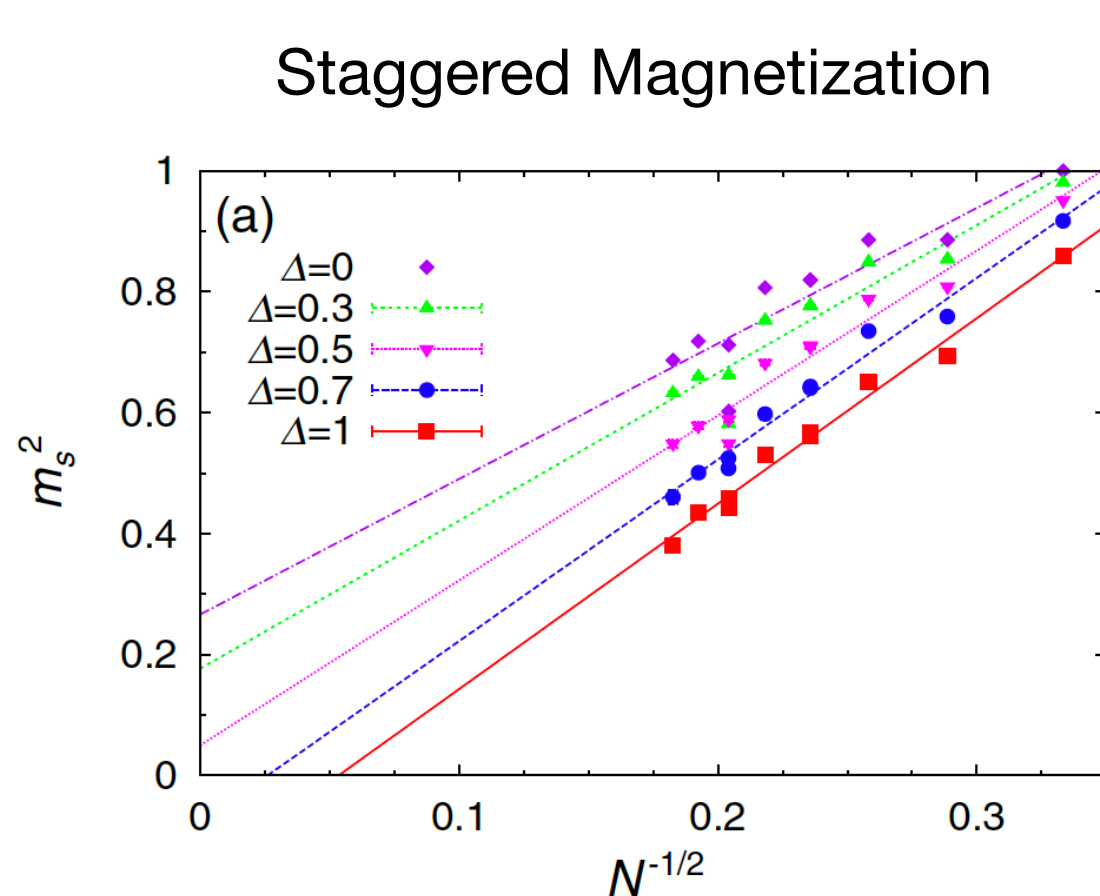
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# Numerical simulation

- ED for a disordered triangular lattice Heisenberg model,

$$H = J \sum_{\langle i,j \rangle} (1 + \Delta \alpha_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j$$

with  $\alpha_{ij}$  drawn from a random uniform distribution,  $\alpha_{ij} \in [-1, 1]$



Extrapolated finite size data signals the destruction of LRO at a finite value of disorder magnitude.

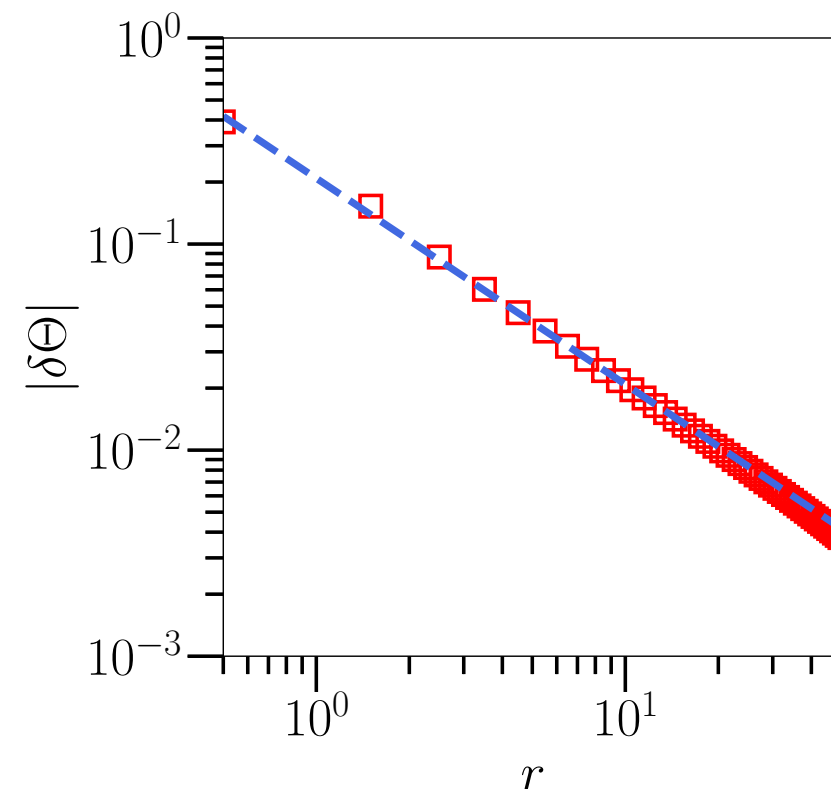
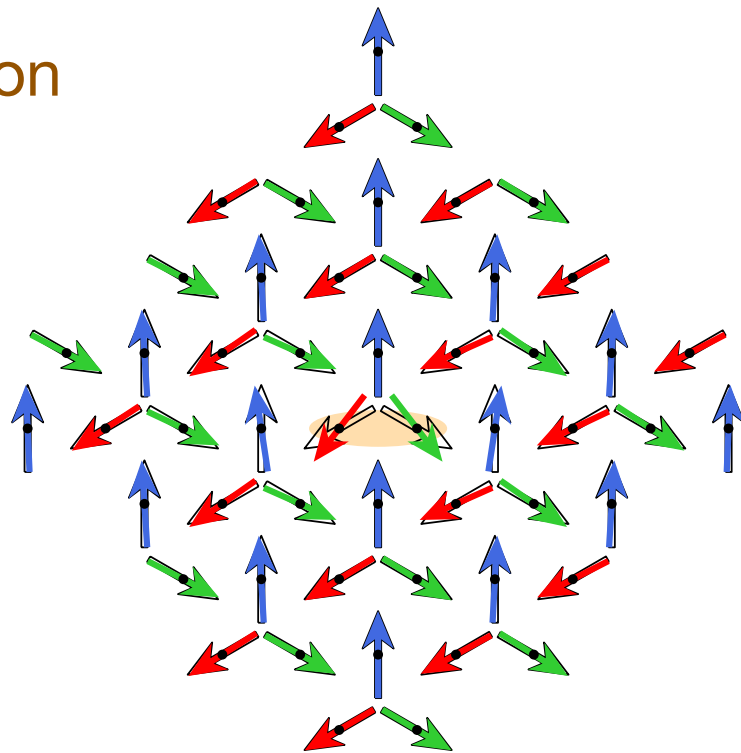
**Isolated bond defect**

# Bond defect induced spin reorientation

- Classical triangular lattice Heisenberg antiferromagnet with bond-defect at the center.

$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad \text{with} \quad J_{ij} = \begin{cases} J - \delta J. & \text{if } (i,j) = (1,0) \\ J, & \text{otherwise} \end{cases}$$

Bond direction  
 $\delta = (1, 0)$



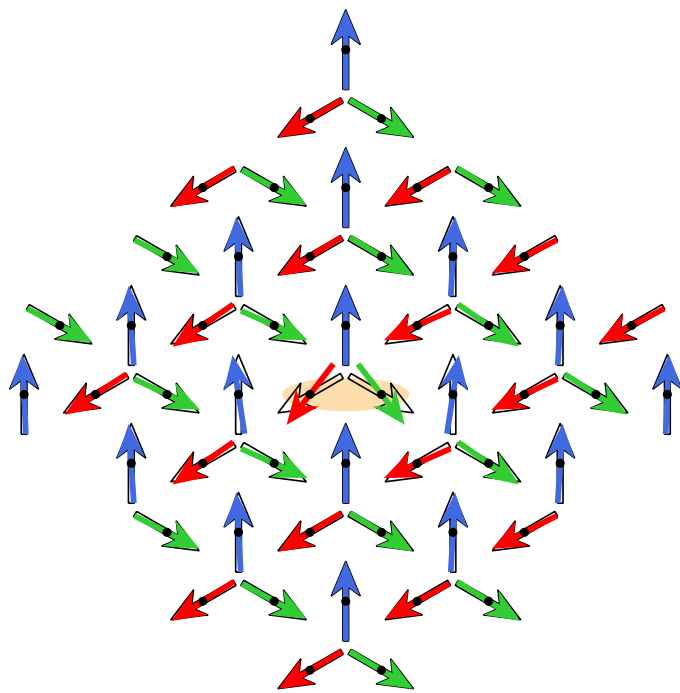
Spins away from the defect are rotated from their 120-degree orientation with,

2D dipolar profile

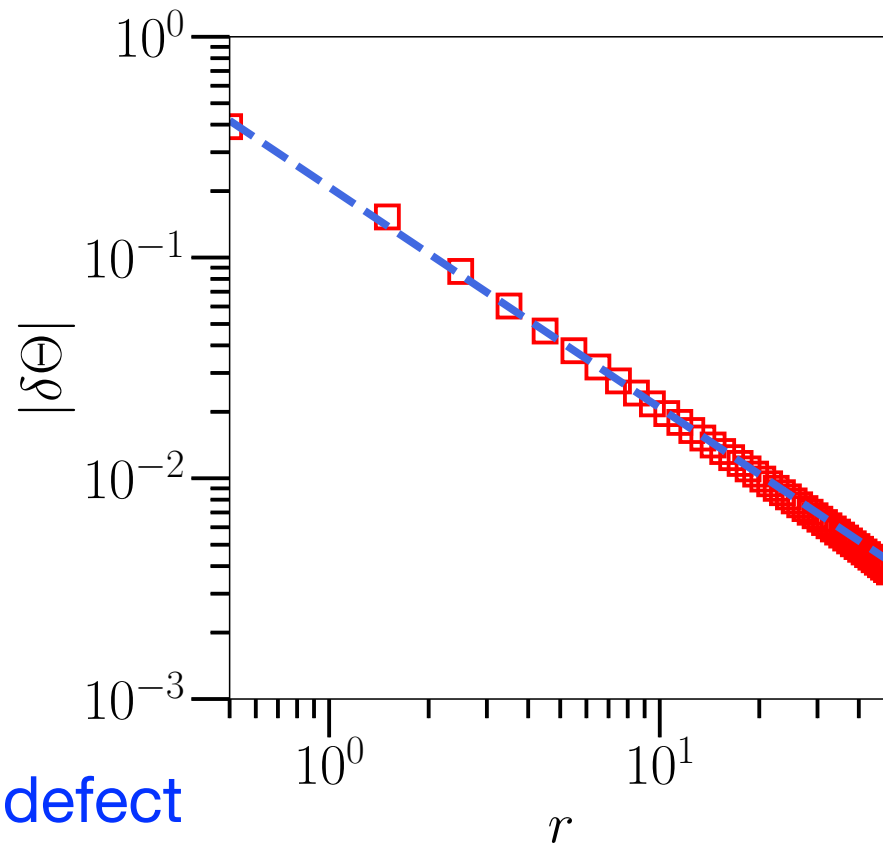
$$\delta\Theta(r) \propto \frac{\cos \theta}{r}$$

Angular dependence dictated by  
dipole moment  $\delta$

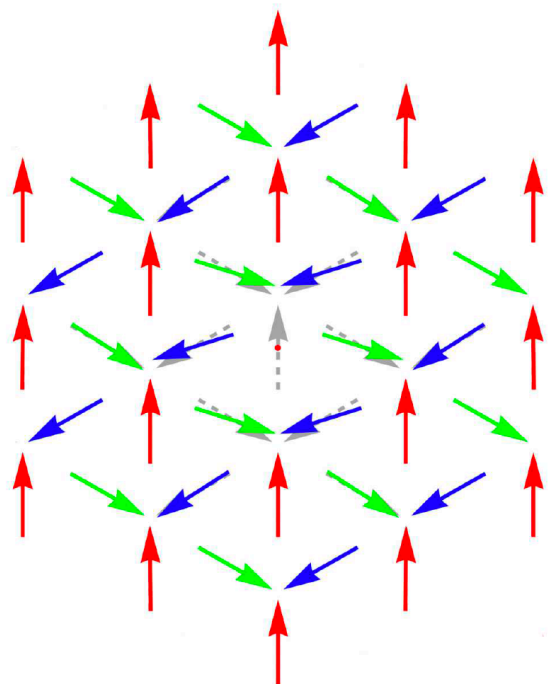
# Bond defect vs vacancy



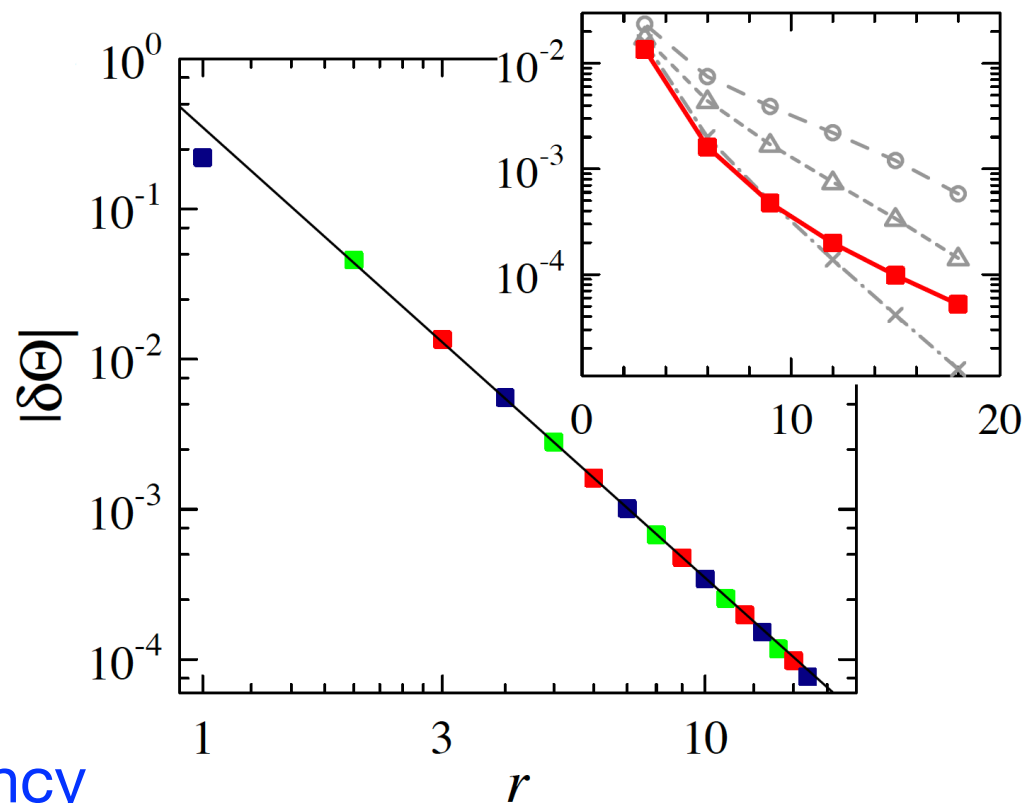
Bond defect



$$\delta\Theta(r) \propto \frac{1}{r}$$



Vacancy



Faster asymptotic decay

$$\delta\Theta(r) \propto \frac{1}{r^3}$$

A. Wollny et. al,  
Phys. Rev. Lett. 107,  
137204 (2011)

# Transverse response

- For a single bond-defect  $\delta J(\mathbf{r}')$  along the bond direction  $\delta(\mathbf{r}')$  the angle correction in an arbitrary point  $\delta\Theta(\mathbf{r})$  is proportional to **transverse fluctuation** of local spins from the ordered state.

$$\langle S_r^\perp \rangle \propto \frac{\delta J(\mathbf{r}')}{J} \sum_{\mathbf{q}} \mathbf{q} \cdot \delta(\mathbf{r}') \frac{e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')}}{q^2}$$

Dipole structure factor  
due to the bond defect

Transverse spin correlation  
 $\langle\langle \mathbf{S}_{\mathbf{q}}^\perp; \mathbf{S}_{-\mathbf{q}}^\perp \rangle\rangle$

Non-collinear  
antiferromagnets with  
spontaneously broken  
SU(2) symmetry.

Isolated bond  
defect



Finite transverse  
fluctuation from the  
ordered state with  
dipolar decay

# Bond disorder

# From bond defect to bond disorder

- For small disorder one can consider superposition of bond defects in the linear response approximation.

$$\langle S_r^\perp \rangle \propto \sum_{\mathbf{q}, \mathbf{r}'} \frac{\delta J(\mathbf{r}')}{J} \frac{\mathbf{q} \cdot \boldsymbol{\delta}(\mathbf{r}')}{q^2} e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')}$$

- For random distribution of bonds:

$$\overline{\delta J(\mathbf{r})} = 0$$

$$\overline{\delta J(\mathbf{r}) \delta J(\mathbf{r}')} = \delta J^2 \delta^{(d)}(\mathbf{r} - \mathbf{r}')$$

$$\langle \overline{S^\perp} \rangle = 0$$

$$\langle \overline{(S^\perp)^2} \rangle = \left( \frac{\delta J}{J} \right)^2 \int \frac{d^d q}{(2\pi)^d} \frac{1}{q}$$



Divergent in  $d = 2$

# Infrared divergence

$$\langle \overline{S^\perp} \rangle = 0$$

$$\langle \overline{(S^\perp)^2} \rangle = \left( \frac{\delta J}{J} \right)^2 \int \frac{d^d q}{(2\pi)^d} \frac{1}{q}$$



Divergent in  $d = 2$

Infrared divergence of transverse  
fluctuation at  $d = 2$  implies  
destruction of LRO

For  $\delta J \rightarrow 0$  this implies an exponential divergence of the  
correlation length.

$$\xi \propto e^{\frac{1}{\delta J/J}}$$

# Contrast to random dilution

Bond defect has slowly decaying transverse response,

$$\langle S^\perp \rangle \propto \frac{1}{r}$$

in  $d = 2$



Concentration of random defects leads to infrared divergence,

$$\overline{\langle (S^\perp)^2 \rangle} \rightarrow \infty$$

LRO is destroyed

Vacancy has a fast decaying transverse response,

$$\langle S^\perp \rangle \propto \frac{1}{r^3}$$

in  $d = 2$

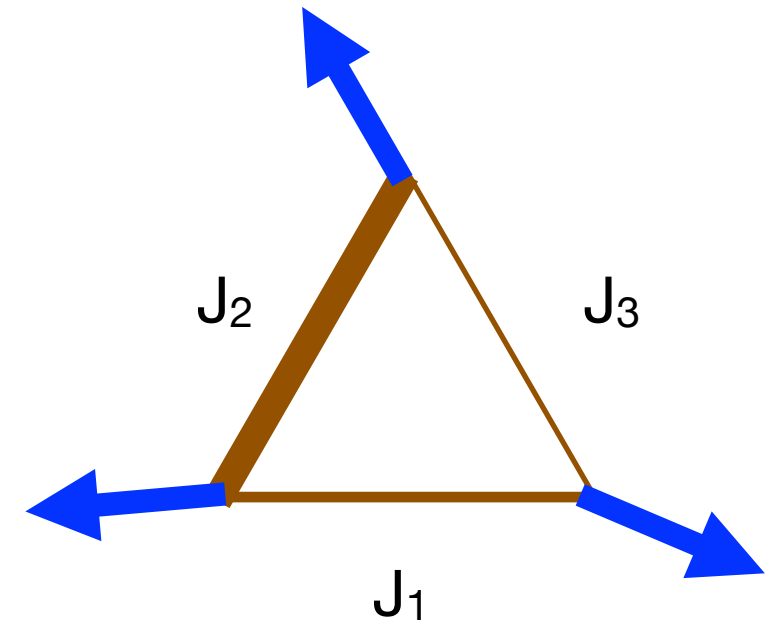
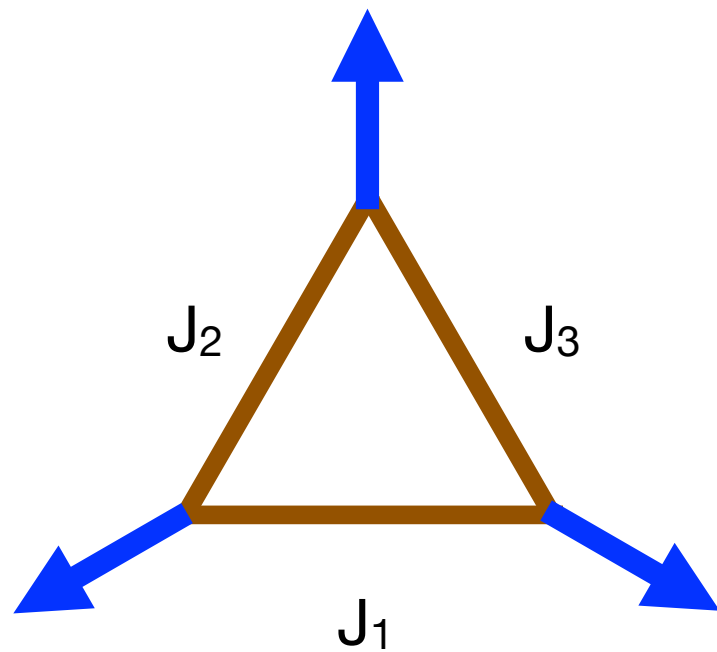


For random concentration of vacancies, i.e. dilution, no infrared divergence of transverse fluctuation.

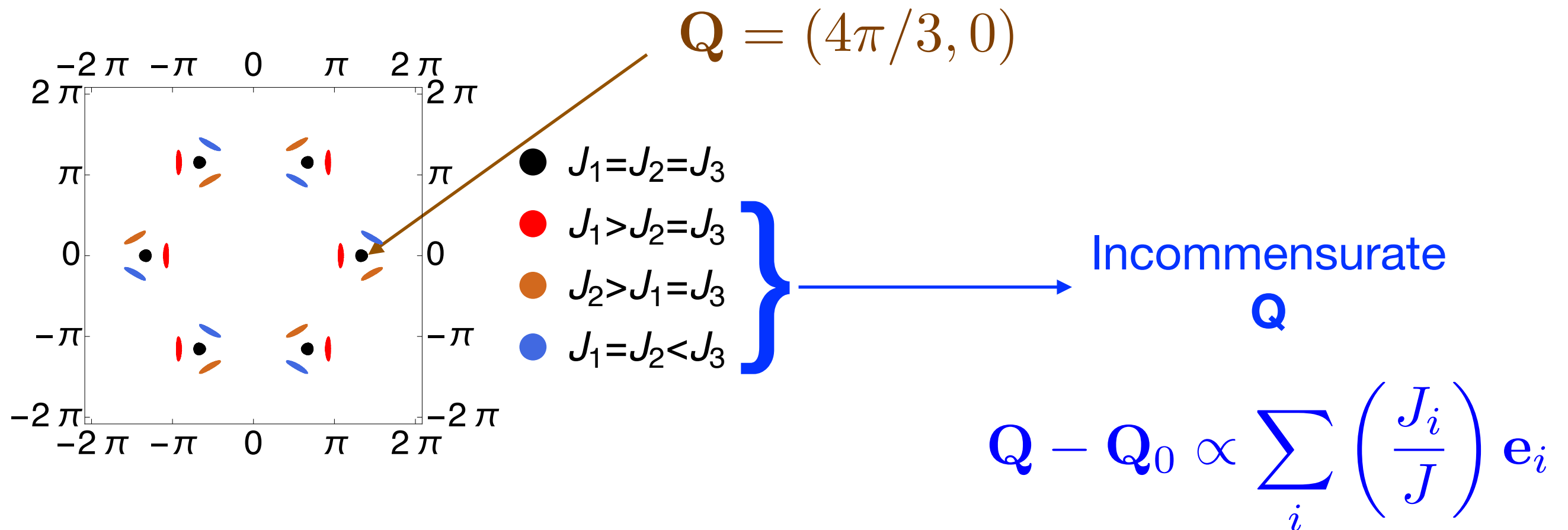
So LRO is stable!

# Effect on the state

Bond disorder turns the triangles locally anisotropic



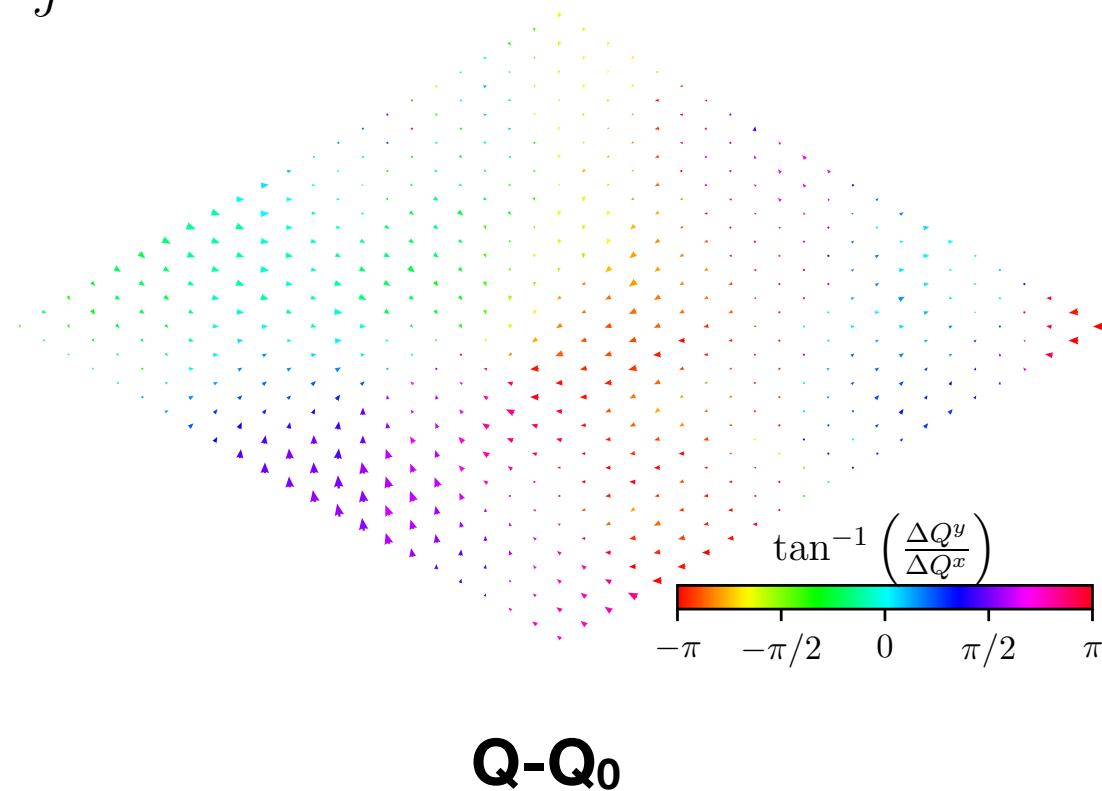
Globally anisotropic triangular lattice



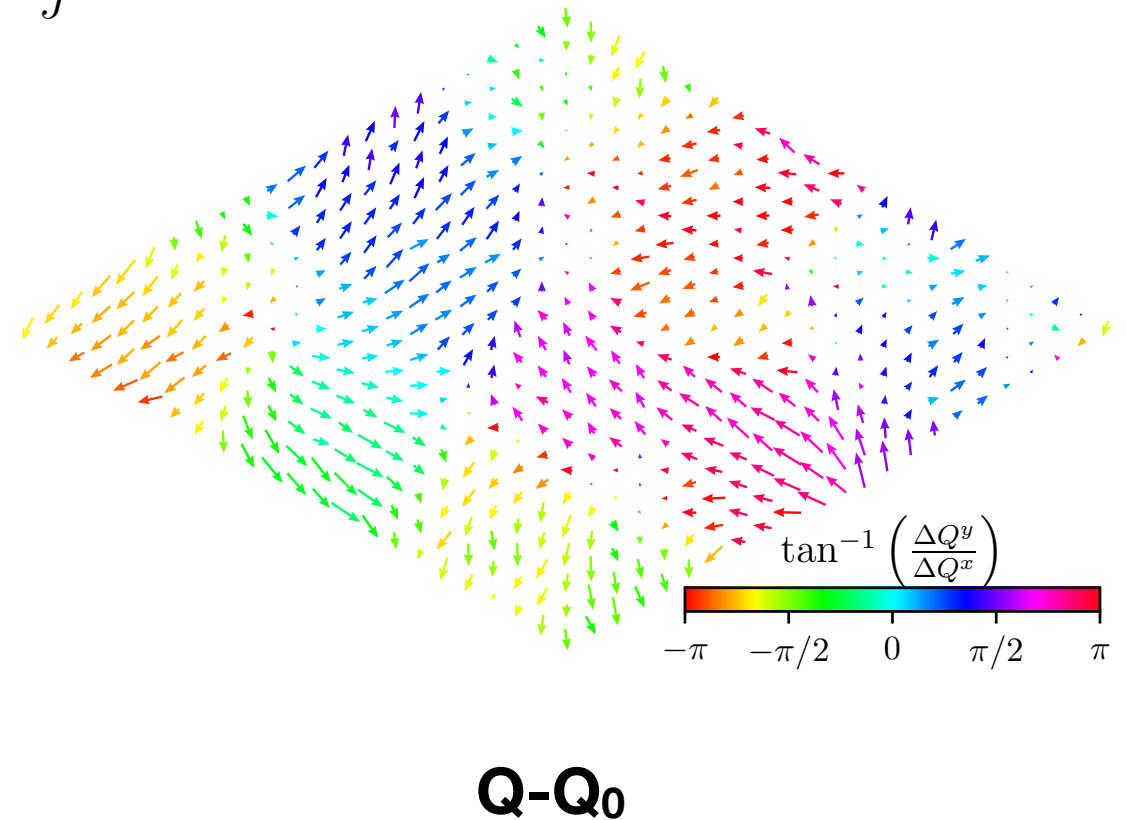
# Numerical simulation

- Glassy state demarcated by domains of incommensurability. **Q-Q<sub>0</sub>**

$$\frac{\delta J}{J}=0.1$$



$$\frac{\delta J}{J}=0.4$$



- Domain length decreases with increasing disorder.

$$\xi \propto e^{\frac{1}{\delta J/J}}$$

# Bond disorder and competing effects

- **Finite temperature:** At finite but low temperature in 2D, thermal fluctuation and bond-disorder effects compete with a crossover temperature of precedence given by,

$$T^* = \left( \frac{\delta J}{J} \right)$$

- **Broken SU(2) symmetry:** The infrared divergence is avoided but but stability of LRO depends on the disorder magnitude and the mass gap, with LRO stable for,

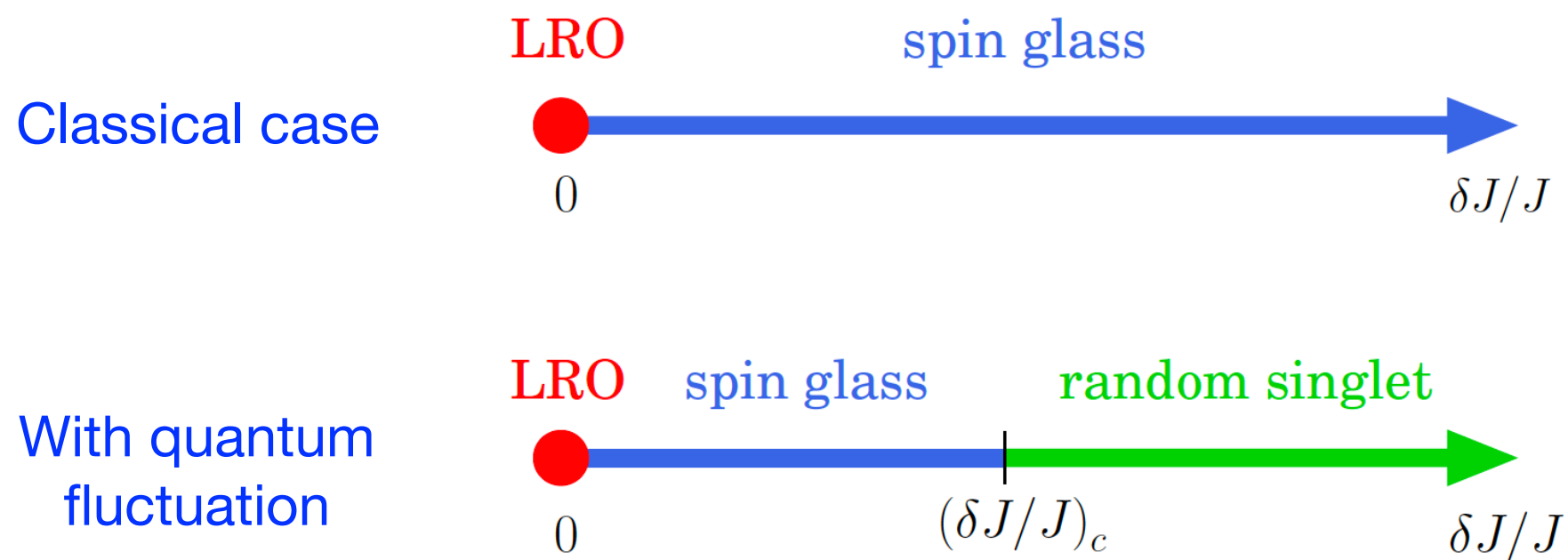
$$m^{2-d} \gg \frac{1}{(2\pi)^{d-1}} \left( \frac{\delta J}{J} \right)^2$$

- **Layered magnet:** The stability of the LRO is dictated by the inter-layer coupling  $J^\perp$  with the stability criterion,

$$\left( \frac{J^\perp}{J} \right) \gg \left( \frac{\delta J}{J} \right)^2$$

# Summary

- Bond defects in non collinear magnets produce dipolar distortion with slow long distance decay ( $\sim \frac{1}{r}$  in  $d = 2$ ).
- In  $d = 2$  or less, random concentration of such defects destroy LRO even for very small disorder magnitude.
- The resulting ground state has short ranged correlation which exponentially diverge for small disorder  $\xi \propto e^{(J/\delta J)}$

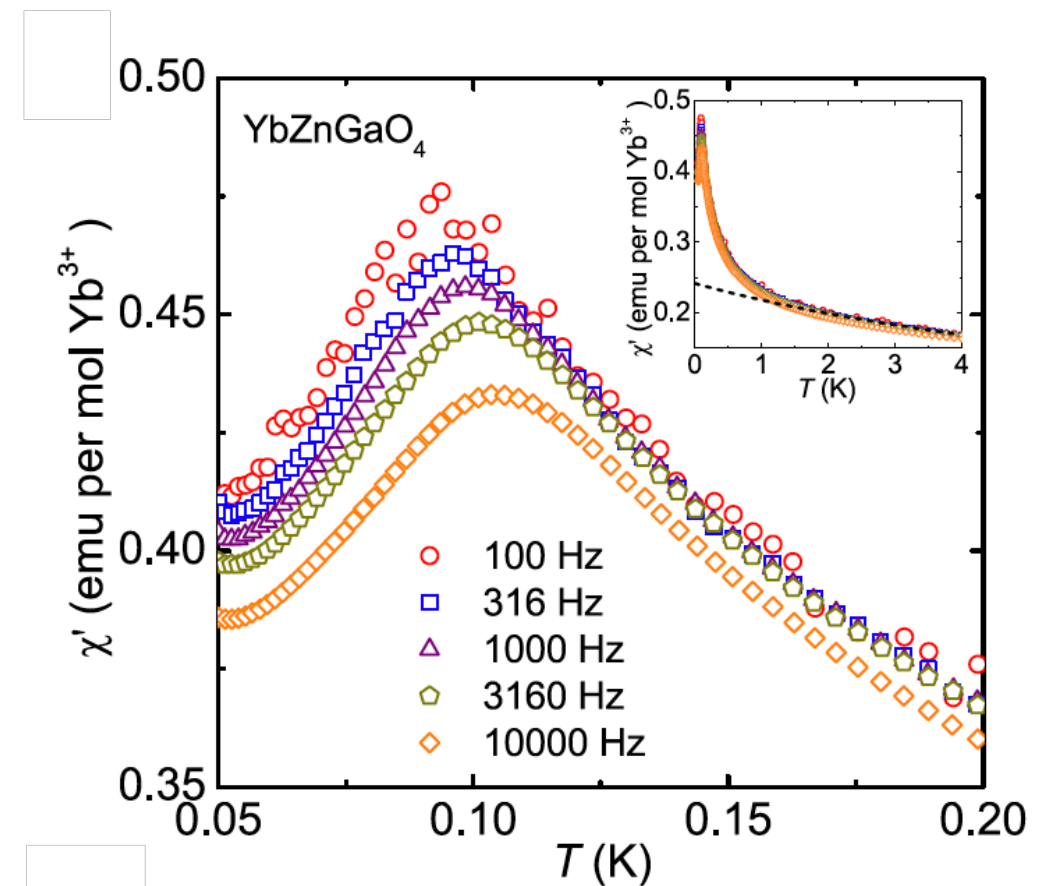
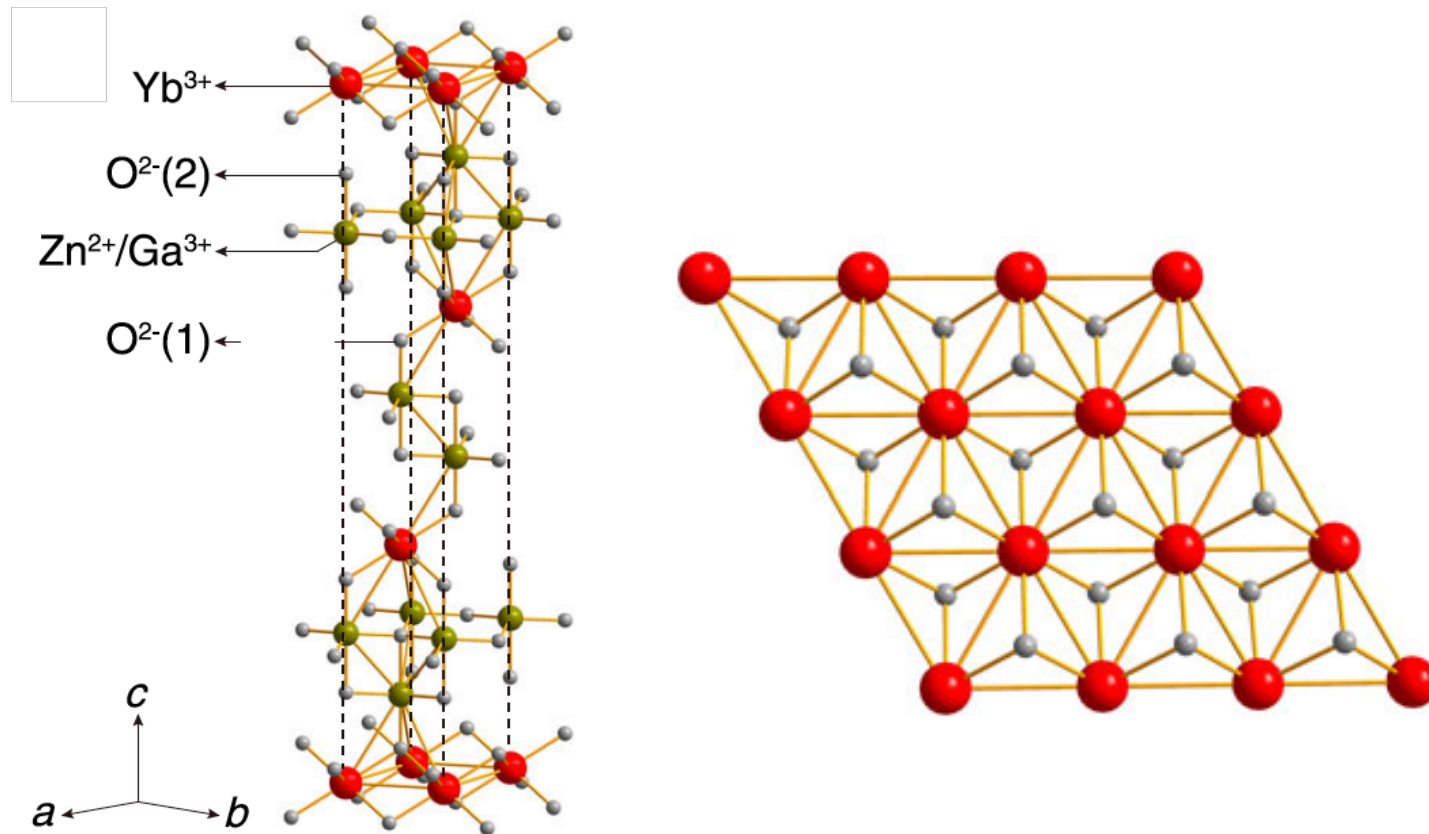


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**Extra slides**

# Experiment

- Experiment on 2D triangular lattice compound  $\text{YbZnGaO}_4$  revealed a spin-glass ground state



Low-frequency response indicating glassy behavior:

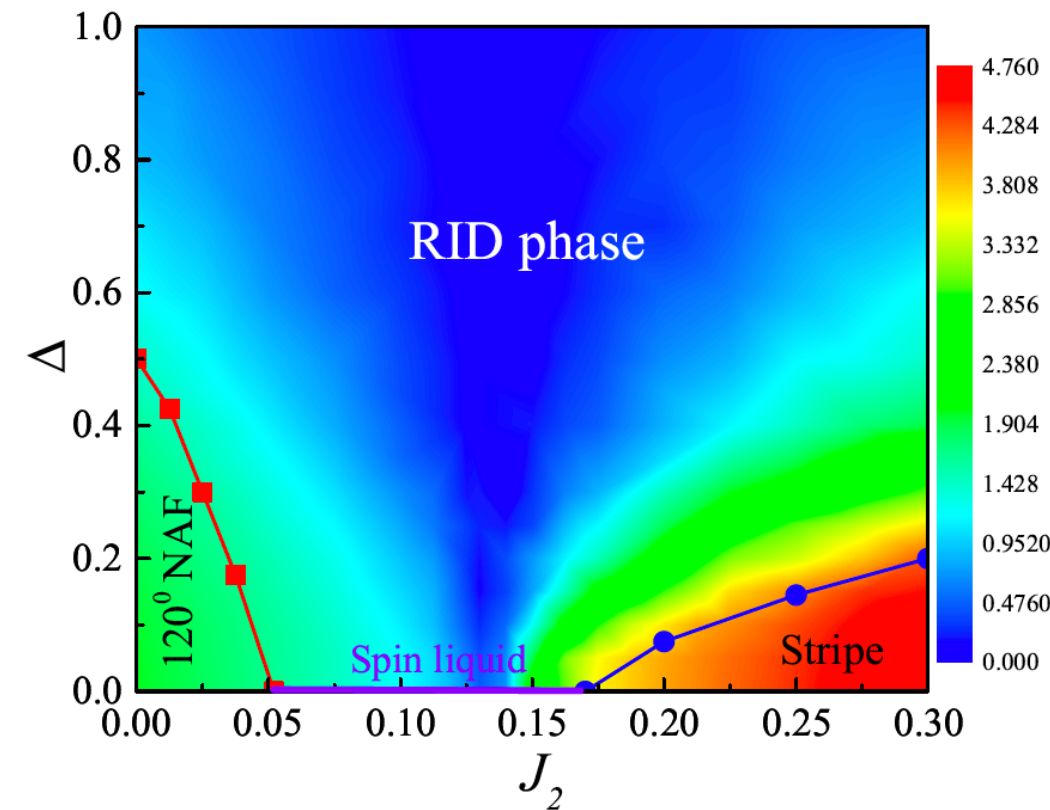
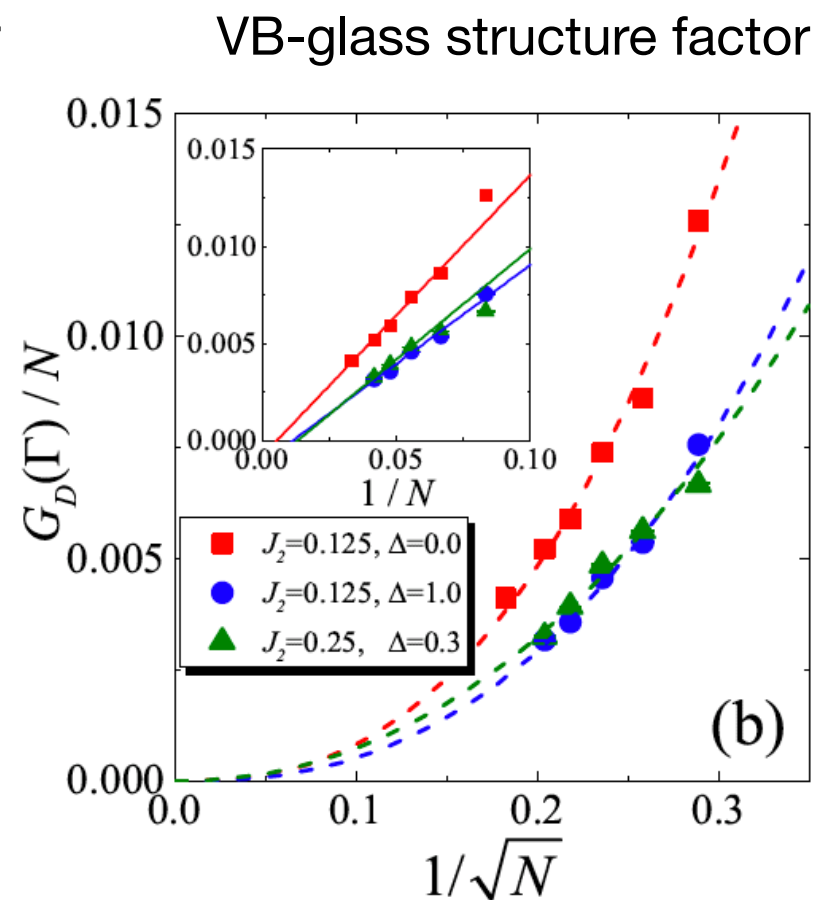
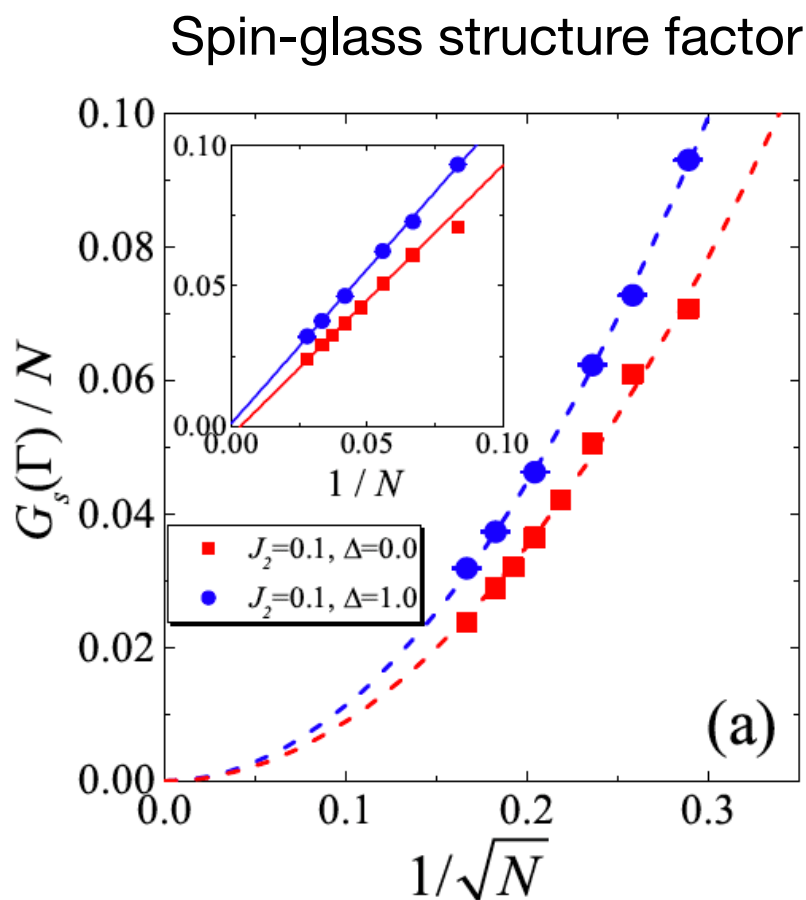
[Z. Ma et. al, Phys. Rev. Lett. 120, 087201 \(2018\)](#)

# Numerical simulation

- ED and DMRG analysis of the disordered  $J_1$ - $J_2$  triangular lattice Heisenberg model,

$$H = J_1 \sum_{\langle i,j \rangle} (1 + \Delta \alpha_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} (1 + \Delta \beta_{ij}) \mathbf{S}_i \cdot \mathbf{S}_j$$

with  $\alpha_{ij}, \beta_{ij} \in [-1, 1]$

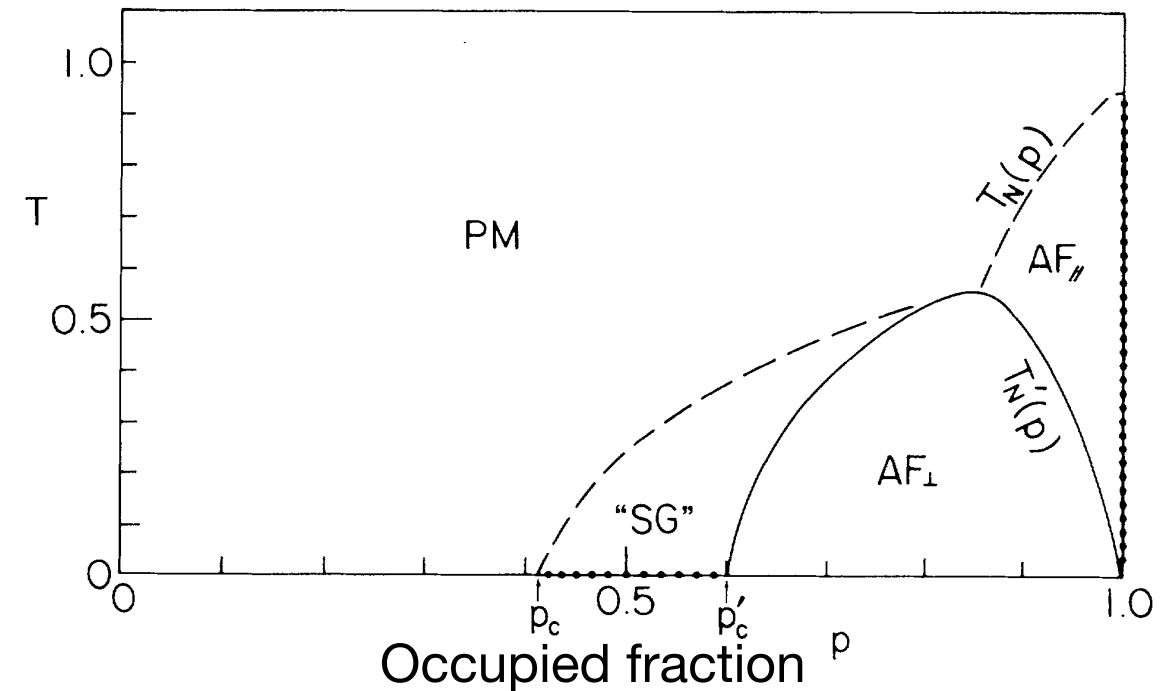


Indication of **Randomness Induced Disordered** phase without magnetic, spin-glass or valence-bond glass ordering

H. Wu et. al, [arXiv:1802.00935](https://arxiv.org/abs/1802.00935) (2018)

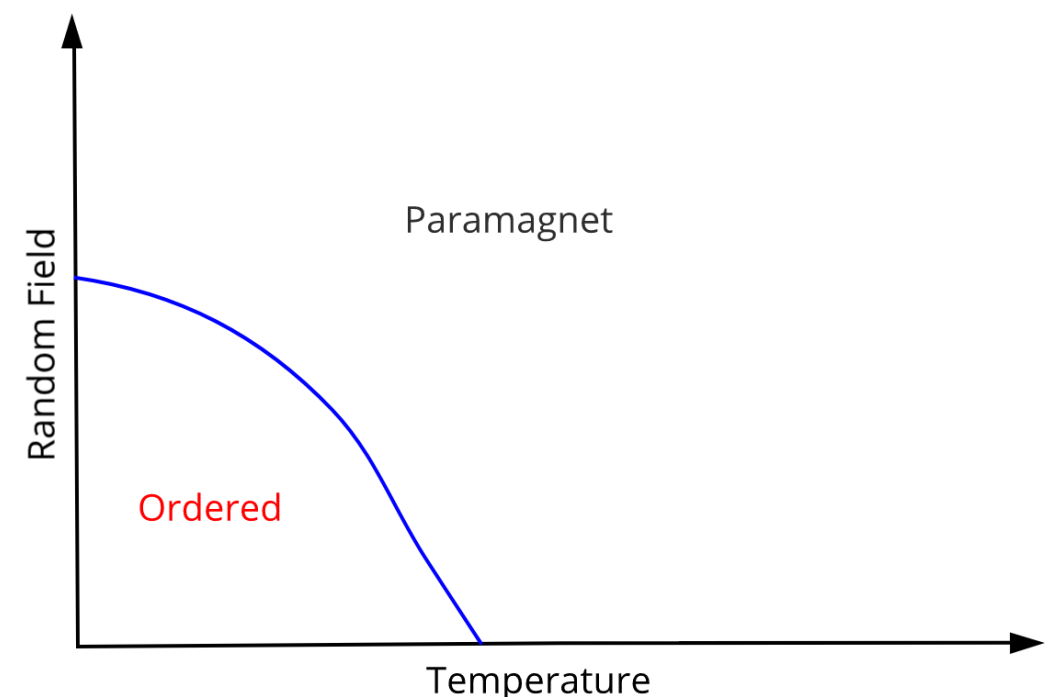
# Quenched disorder affecting long range order

- Random site-dilution can induce glassy behavior or long range magnetic order depending on competing interactions of a model.



The phase diagram of site diluted  $J_1$ - $J_2$  XY antiferromagnet:  
[C. L. Henley, Phys. Rev. Lett. 62, 2056 \(1989\)](#)

- Infinitesimal random fields destroy long range order (LRO) below or at the lower critical dimension
- Imry-Ma argument:**  $d_c = 2$  for discrete symmetries and  $d_c = 4$  for continuous symmetries



The theoretical phase diagram of 3D random-field Ising antiferromagnet